Deep Learning Baseline Comparison Plan

WVF/LF vs Learned Edge Detectors Across Noise Regimes

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1. Objective

Determine at what noise level, if any, the training-free WVF/LF overtakes state-of-the-art deep learning edge detectors. This directly tests Bagan's central thesis that classical image processing methods are more reliable than ML-based approaches in degraded imaging conditions.

2. Available Models

Seven pre-trained deep learning edge detectors are available. Table 1 summarizes their architecture, size, training data, and readiness.

Model	Architecture	Params	Trained on	Venue	Status
DexiNed	Dense Inception CNN	12M	BIPED	WACV 2020	Ready
TEED	Lightweight CNN	58K	BIPED	ICCV-W 2023	Ready
pidinet	Pixel Difference CNN	700K	BSDS500	ICCV 2021	Ready
EDTER	Vision Transformer	25M	BSDS500	CVPR 2022	Needs setup
EdgeNAT	Neighborhood Attn. Trans.	varies	BSDS500	—	Needs setup
NBED	CAFormer Enc-Dec	8M	BSDS500	—	Mostly ready
DiffusionEdge	Latent Diffusion	100M	BIPED/BSDS	AAAI 2024	Ready

Table 1: Available deep learning edge detectors. "Ready" means pre-trained weights are included and inference scripts exist. Models trained on BIPED may show domain shift on BSDS500/UDED and vice versa.

3. Expected Performance by Noise Regime

Table 2 summarizes predictions based on each model's architecture and training.

Model	Clean	Moderate noise (SNR 2–4)	Heavy noise (SNR < 1)
DexiNed	Strong on BIPED, weaker OOD	Moderate degradation	Likely fails — no noise augmentation in training
TEED	Good, fast	Degrades faster (tiny model)	Poor — too few params to denoise
pidinet	Strong on BSDS500	Moderate	Moderate — pixel differences may resist some noise
EDTER	Best on clean (transformer)	Degrades gradually	Poor — fixed receptive field
DiffusionEdge	Strong	May resist well	Best DL candidate — diffusion is inherently denoising
WVF (optimal)	0.84 ODS	Competitive at Np=100+	Wins at Np=250–500 with d=2
LF (optimal)	0.84 ODS	Wins with m=5–10	Wins with m=14–20

Table 2: Expected performance trends. DL models were trained on clean data and are expected to degrade at low SNR. WVF/LF with large support should eventually overtake them. DiffusionEdge is the most interesting comparison because diffusion models have built-in denoising.

The key prediction: there exists an SNR crossover below which WVF/LF with appropriate parameters outperforms every DL model. If this crossover is at SNR $\approx$ 1–2, Bagan's argument has merit for underwater/sonar applications. If the DL models remain superior even at SNR = 0.3, the argument does not hold.

4. Study Design

4.1. Phase 1: Ready Models (4 models, 6 GPU-hours)

Run the four ready-to-use models (DexiNed, TEED, pidinet, DiffusionEdge) on the same 20 images $\times$ 8 SNR levels $\times$ 5 noise types from the WVF/LF noise ablation. Each model produces a single edge probability map per image — no parameter grid needed.

Model	Per-image time	Images	Conditions	Total
DexiNed	≈ 0.5 s	20	36	≈ 6 min
TEED	≈ 0.05 s	20	36	≈ 0.6 min
pidinet	≈ 0.1 s	20	36	≈ 1.2 min
DiffusionEdge	≈ 5 s	20	36	≈ 60 min
Total inference				**$\approx$ 68 min**
Evaluation (1001 thresh)				≈ 15 min
Phase 1 total				**$\approx$ 1.5 h**

Table 3: Phase 1 runtime estimate. DiffusionEdge dominates due to iterative denoising (configurable sampling steps). All other models are fast single-forward-pass networks.

4.2. Phase 2: Setup Models (2–3 models, 4 GPU-hours)

Set up EDTER, EdgeNAT, and NBED by downloading pre-trained weights and configuring the MMSegmentation environment. Then run the same evaluation.

EDTER and EdgeNAT require the MMSegmentation framework (MMCV 1.2.2, specific PyTorch versions). This may require a separate conda environment. Estimated setup time: 1–2 hours.

4.3. Phase 3: Analysis (1 hour)

Generate comparison plots:

- ODS vs SNR curves: one line per model plus WVF/LF best, colored by method type (classical vs learned)
- Crossover identification: at which SNR does the best WVF/LF config beat each DL model?
- Noise-type sensitivity: which models are robust to speckle vs Gaussian vs salt-and-pepper?
- Computational cost: ODS vs FLOP scatter including DL model inference cost

5. SLURM Configuration

The models can be run in parallel since they don't interfere with each other. Each model gets its own SLURM job.

Job	Partition	GPUs	Time	Memory
DexiNed + TEED + pidinet	gpu-A100	1	1 hour	32 GB
DiffusionEdge	gpu-A100	1	2 hours	64 GB
EDTER	gpu-A100	1	2 hours	64 GB
EdgeNAT	gpu-A100	1	2 hours	64 GB
NBED	gpu-A100	1	1 hour	32 GB

Table 4: Recommended SLURM configuration. The three lightweight models (DexiNed, TEED, pidinet) can share a single job since their combined runtime is under 10 minutes. DiffusionEdge needs its own job due to higher memory and compute. The MMSegmentation models (EDTER, EdgeNAT) each need separate jobs with their own environment setup.

All jobs can be submitted simultaneously with `sbatch`. The A100 partition has 6 mix-state nodes plus 1 idle node, so 5 parallel jobs should schedule without waiting. Use gpu-A100 instead of gpu-H200 to avoid competing with the running noise ablation job.

6. Key Comparisons

The final analysis should produce three main results.

Result 1: The crossover SNR. A table showing, for each DL model and noise type, the SNR below which the best WVF/LF configuration achieves higher ODS. If this varies by noise type (e.g., crossover at SNR = 2 for Gaussian but SNR = 4 for speckle), that is a significant finding.

Result 2: Cost-normalized performance. DL models are expensive (DiffusionEdge: 1.2 GB weights, iterative sampling). WVF with $N_p = 25$, $d = 2$, $N_s = 3$ uses 0.1 GFLOP per image. If WVF matches DL performance at moderate noise while using 1000× less compute, that is a practical advantage for edge/embedded deployment — exactly the scenario Bagan argues for.

Result 3: Domain shift vs noise robustness. DexiNed and TEED are trained on BIPED. How do they perform on UDED (underwater, their target domain but out of training distribution) vs BSDS500 (natural scenes, also OOD)? The WVF/LF is training-free and domain-agnostic. If DL models fail on UDED even at clean SNR due to domain shift, that validates the WVF/LF approach for novel domains.